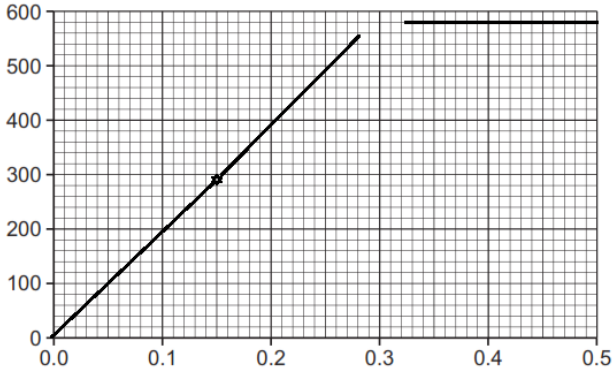


2(a)	extracellular (enzyme) ;	1
2(b)(i)	mmol dm⁻³ ; curve drawn, which starts at 0.0 <u>and</u> passes through at y-axis 290 U cm⁻³ and x-axis 0.15dm⁻³ and plateau is at 580 U cm⁻³ ; 	2

Question	Answer	Marks																
2(b)(ii)	can gain max from agree, or disagree, or both any three from: agree Figure 2.1, substrate / cholesterol, concentration 1 can use, known cholesterol concentrations / standards, to produce a calibration curve ; 2 enzyme is active at (substrate) concentrations, tested / over a range / at least to 0.5 mmol dm⁻³ ; 3 suggestion that, low K_m / K_m of 0.15 mmol dm⁻³, allows enzyme to react with low concentrations / affinity of enzyme to substrate is high ; AW Figure 2.2, pH 4 enzyme (quite) active, qualified ; e.g. at a (fairly) wide range of pH at a number of different pHs between pH 6 and pH9 in acidic and alkaline conditions <i>in context of pH 6–9</i> I because most foods are at, optimum pH for enzyme / pH7–8 5 use of data from Figure 2.2 to support ; ecf on I in mp4 <i>ref. to optimum between pH7 99% and pH8 98%</i> <table><tr><td>pH</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>% max</td><td>7</td><td>11</td><td>21</td><td>99</td><td>98</td><td>57</td><td>8</td></tr></table> general 6 AVP ; e.g.bacterial enzyme so, easy / cheap, to obtain bacterial enzyme so can be obtained in large quantities disagree Figure 2.1, substrate / cholesterol, concentration 7 concentrations higher than 0.5 mmol dm⁻³, not tested ;	pH	4	5	6	7	8	9	10	% max	7	11	21	99	98	57	8	3
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% max	7	11	21	99	98	57	8											

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2(b)(ii)	8 concentrations higher than 0.5 mmol dm^{-3}, may inhibit activity ; <i>Figure 2.2, pH</i> 9 enzyme not very active at pHs below / above / either side of optimum / either side of pH 7– pH 8 / 9 ; AW e.g. idea that good enzyme activity limited to, pH 7–8 / pH 7– 8 / 9 10 use of data from Figure 2.2 to support ; <i>use of data once only</i> <table><tr><td>pH</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>% max</td><td>7</td><td>11</td><td>21</td><td>99</td><td>98</td><td>57</td><td>8</td></tr></table> 11 <i>idea that</i> (many) foods, can have lower or higher pH / are too acidic or alkaline, for enzyme activity needed ; A basic for alkaline <i>general</i> 12 AVP ; e.g. temperature used (to obtain results) may not be suitable for food industry testing conditions only shows relative activity so activity at lower pHs may be poor	pH	4	5	6	7	8	9	10	% max	7	11	21	99	98	57	8	
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Question	Answer	Marks
2(c)	<i>context can be enzyme on pathogen surface or secreted enzyme</i> <i>max 2 if no ref. to entry</i> any three from: <i>context is cell surface membrane of host cell</i> (membrane) cholesterol, broken down / decreases / AW ; (membrane) becomes, less stable / more permeable / more fluid ; <i>ref. to, gaps / spaces, created (between phospholipids) for (easier entry of), bacterial cell / M. tuberculosis ;</i> pores easier for / AW, endocytosis of bacteria to occur / endocytotic vesicles with bacteria to form ; reduces cholesterol content needed to, form / maintain, (cell) membranes ; transport of substances across membranes affected (because of damage to membrane(s)) ; A example(s) impairs / AW, organelle function (because of damage to membrane(s)) ; A example(s) other example of harm to the cell ; e.g. products of reaction catalysed by enzyme may be toxic to cell <i>M. tuberculosis</i> / AW, releases toxins <i>M. tuberculosis</i> / AW, uses host, cell nutrients / respiratory substrates / ATP damages organelle membranes AVP ; e.g. less cholesterol available for host cell to produce other lipids	3

Question	Answer	Marks														
2(d)	<i>allow AW throughout</i> <i>allow bacteria / pathogen, for M. tuberculosis</i> <i>for each mark, needs to have column 1 and column 2</i> <i>any three from:</i> <table><tr><th>column 1</th><th>column 2</th></tr><tr><td>1 slows / stops, enzyme action A enzyme cannot, bind / breakdown, cholesterol</td><td>(drug) acts as an inhibitor (competitive / non-competitive) ; <i>if term not used allow a correct description</i></td></tr><tr><td>2 won't harm, human cells A body / tissues / organs</td><td>only acts on pathogen / enzyme not found in human cells or works (only) on cholesterol oxidase;</td></tr><tr><td>3 processes (in host cell) unaffected</td><td></td></tr><tr><td>4 immune system more likely to be effective / described</td><td>bacteria, remain outside / not inside / prevented entering, cells ; bod difficult to enter</td></tr><tr><td>5 reduces problems of developing antibiotic resistance A does not / less likely to, develop resistance to, drug / inhibitor</td><td>can use, instead of / with / fewer, antibiotics A can just take the drug or bacteria susceptible to drug ; <i>allow this idea if mp1 gained</i></td></tr><tr><td>6 use for, TB caused by drug-resistant bacteria / named e.g. MDR (multi) / XDR (extensively) / TDR (total) / PDR (pan)</td><td></td></tr></table>	column 1	column 2	1 slows / stops, enzyme action A enzyme cannot, bind / breakdown, cholesterol	(drug) acts as an inhibitor (competitive / non-competitive) ; <i>if term not used allow a correct description</i>	2 won't harm, human cells A body / tissues / organs	only acts on pathogen / enzyme not found in human cells or works (only) on cholesterol oxidase;	3 processes (in host cell) unaffected		4 immune system more likely to be effective / described	bacteria, remain outside / not inside / prevented entering, cells ; bod difficult to enter	5 reduces problems of developing antibiotic resistance A does not / less likely to, develop resistance to, drug / inhibitor	can use, instead of / with / fewer, antibiotics A can just take the drug or bacteria susceptible to drug ; <i>allow this idea if mp1 gained</i>	6 use for, TB caused by drug-resistant bacteria / named e.g. MDR (multi) / XDR (extensively) / TDR (total) / PDR (pan)		3
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Question	Answer		Marks
2(d)	column 1	column 2	
	7 prevents, latent TB / dormant TB / <i>M. tuberculosis</i> staying alive in host cells	prevents entry (to cells) ; AW bod difficult to enter <i>if not stated, allow one mark for any two of 7,8,9 in column 1</i>	
	8 stops, reproduction (in cells) / replication R mitosis A ref. to binary fission		
	9 decreases, number of bacteria / chance of transmission A helps, break the transmission cycle / prevent the spread of (antibiotic) resistant bacteria		
	10 few / no, side-effects	idea of treatment more, acceptable / efficient (compared to other treatments)	
	11 treatment duration may be short	allow if column 1 uses comparative phraseology	